

A NEW FORMULA FOR SUMS OF POWERS USING CENTRAL NEWTON'S INTERPOLATION

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ABSTRACT.

CONTENTS

Preliminaries	2
1. Definitions and notations	2
2. Historical context	2
3. Central Newton's formula	2
4. Helper lemmas	3
5. Ordinary sums of powers	5
6. Remark about central factorial numbers of the second kind	6
7. Generalized central factorial numbers of the second kind	6
8. Double and triple sums of powers in terms of central differences	8
9. Multifold sums of powers in terms of central differences	10
10. Multifold sums of powers in generalized central factorial numbers	10
11. Multifold sums of powers in binomial form	12
12. Related works	13
13. Framework for sums of powers	13
14. Conclusions	13
References	13
15. Mathematica programs	13

PRELIMINARIES

Lemma 0.1 (Stirling form of Binomial). *For integers n, k*

$$\binom{n}{k} = \sum_{j=0}^k \frac{(-1)^{k-j}}{k!} \begin{bmatrix} k \\ j \end{bmatrix} n^j$$

1. DEFINITIONS AND NOTATIONS

Allow us to define the mathematical objects we utilize throughout the manuscript. For multifold sums of powers, we use the notation provided by Donald Knuth in [1], because it is simple and beautiful.

Definition 1.1 (Multifold sums of powers). *For non-negative integers r, n, m*

$$\Sigma^0 n^m = n^m,$$

$$\Sigma^1 n^m = \Sigma^0 1^m + \Sigma^0 2^m + \cdots + \Sigma^0 n^m,$$

$$\Sigma^{r+1} n^m = \Sigma^r 1^m + \Sigma^r 2^m + \cdots + \Sigma^r n^m.$$

We utilize the following notation for central factorials, see also [2, 3, 4]

Definition 1.2 (Central factorials). *For integers n, k*

$$n^{[k]} = \begin{cases} 0, & \text{if } k < 0, \\ 1, & \text{if } k = 0, \\ n \left(n + \frac{k}{2} - 1\right) \left(n + \frac{k}{2} - 2\right) \cdots \left(n - \frac{k}{2} + 1\right) = n \left(n + \frac{k}{2} - 1\right)_{k-1}, & \text{if } k > 0, \end{cases}$$

where $\left(n + \frac{k}{2} - 1\right)_{k-1} = \prod_{j=1}^{k-1} \left(n + \frac{k}{2} - j\right)$ are falling factorials.

2. HISTORICAL CONTEXT

3. CENTRAL NEWTON'S FORMULA

Proposition 3.1. (Central Newton's formula [2, p. 462]). *For arbitrary integers x , and t*

$$f(x) = \sum_{k=0}^{\infty} \frac{(x-t)^{[k]}}{k!} \delta^k f(t),$$

where $\delta^k f(t)$ denotes central finite difference

$$\delta^k f(t) = \sum_{j=0}^k (-1)^j \binom{k}{j} f(t + \frac{k}{2} - j),$$

and $(x - t)^{[k]}$ are central factorials defined by (1.2).

4. HELPER LEMMAS

Lemma 4.1 (Binomial form of central factorials). *For integers n and $k \geq 1$,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k - 1}.$$

Proof. We have

$$\frac{n^{[k]}}{k!} = \frac{n}{k!} \left(n + \frac{k}{2} - 1 \right)_{k-1} = \frac{n}{k(k-1)!} \left(n + \frac{k}{2} - 1 \right)_{k-1} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k - 1},$$

because of the identity in falling factorials $\frac{(x)_n}{n!} = \binom{x}{n}$. □

Lemma 4.2 (Halved central factorials). *For integers $n, k \geq 0$,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k - 1} = \frac{1}{2} \binom{n + \frac{k}{2}}{k} + \frac{1}{2} \binom{n + \frac{k}{2} - 1}{k}.$$

Proof. We have $\binom{a}{k} = \frac{a}{k} \binom{a-1}{k-1}$. Let $a = n + \frac{k}{2}$, then

$$\binom{n + \frac{k}{2}}{k} = \frac{n + \frac{k}{2}}{k} \binom{n + \frac{k}{2} - 1}{k-1} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} + \frac{1}{2} \binom{n + \frac{k}{2} - 1}{k-1}.$$

Thus,

$$\frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} = \binom{n + \frac{k}{2}}{k} - \frac{1}{2} \binom{n + \frac{k}{2} - 1}{k-1}.$$

By moving under common denominator, and applying binomial recurrence yields

$$\frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} = \frac{1}{2} \left[2 \binom{n + \frac{k}{2}}{k} - \binom{n + \frac{k}{2} - 1}{k-1} \right] = \frac{1}{2} \left[\binom{n + \frac{k}{2}}{k} + \binom{n + \frac{k}{2} - 1}{k} + \binom{n + \frac{k}{2} - 1}{k-1} - \binom{n + \frac{k}{2} - 1}{k-1} \right].$$

Thus, the statement follows. □

Proposition 4.3 (Central Newton's formula for powers). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$n^m = \sum_{k=0}^m \frac{(n-t)^{[k]}}{k!} \delta^k t^m.$$

Proposition 4.4 (Halved central factorial powers). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$n^m = \frac{1}{2} \sum_{k=0}^m \left[\binom{n-t+\frac{k}{2}}{k} + \binom{n-t+\frac{k}{2}-1}{k} \right] \delta^k t^m$$

Proposition 4.5 (Generalized hockey-stick identity). *For integers a, b and j ,*

$$\sum_{k=a}^b \binom{k}{j} = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

Proof. We have, $\sum_{k=a}^b \binom{k}{j} = \binom{a}{j} + \binom{a+1}{j} + \cdots + \binom{b}{j} = \left(\sum_{k=0}^b \binom{k}{j} \right) - \left(\sum_{k=0}^{a-1} \binom{k}{j} \right)$. By hockey stick identity $\sum_{k=0}^n \binom{k}{j} = \binom{n+1}{j+1}$ yields,

$$\sum_{k=a}^b \binom{k}{j} = \left(\sum_{k=0}^b \binom{k}{j} \right) - \left(\sum_{k=0}^{a-1} \binom{k}{j} \right) = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

This completes the proof. □

Lemma 4.6 (Centered hockey-stick identity). *For integers $n \geq 1$, and arbitrary integers t, k, r*

$$\sum_{j=1}^n \binom{j-t+\frac{k}{2}+r}{k} = \binom{n-t+\frac{k}{2}+r+1}{k+1} - \binom{1-t+\frac{k}{2}+r}{k+1}.$$

Proof. By setting $j = 1 - t + \frac{k}{2} + r$, and $b = n - t + \frac{k}{2} + r$ yields

$$\sum_{j=1}^n \binom{j-t+\frac{k}{2}+r}{k} = \sum_{j=1-t+\frac{k}{2}+r}^{n-t+\frac{k}{2}+r} \binom{j}{k}.$$

By Generalized hockey-stick identity (4.5),

$$\sum_{j=1-t+\frac{k}{2}+r}^{n-t+\frac{k}{2}+r} \binom{j}{k} = \binom{n-t+\frac{k}{2}+r+1}{k+1} - \binom{1-t+\frac{k}{2}+r}{k+1}.$$

Thus, the claim follows. □

5. ORDINARY SUMS OF POWERS

Proposition 5.1 (Ordinary sums of powers). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^m \delta^k t^m \sum_{j=1}^n \left[\binom{j-t+\frac{k}{2}}{k} + \binom{j-t+\frac{k}{2}-1}{k} \right]$$

Proposition 5.2 (Ordinary sums of powers decomposition). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^m \left[\binom{n-t+\frac{k}{2}+1}{k+1} - \binom{1-t+\frac{k}{2}}{k+1} + \binom{n-t+\frac{k}{2}}{k+1} - \binom{-t+\frac{k}{2}}{k+1} \right] \delta^k t^m$$

Proposition 5.3 (Central factorial numbers sums). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^m \left[\binom{n+\frac{k}{2}+1}{k+1} - \binom{1+\frac{k}{2}}{k+1} + \binom{n+\frac{k}{2}}{k+1} - \binom{\frac{k}{2}}{k+1} \right] k! T(m, k),$$

where $T(n, k) = \frac{\delta^{k0} n^m}{k!}$ denote Central factorial numbers of the second kind (in Riordan's notation, see [3, p. 213], and [4].), defined by central Newton's formula for powers, evaluated in $t = 0$

$$n^m = \sum_{k=0}^{\infty} T(m, k) n^{[k]}.$$

Example 5.4 (Sums of squares in central differences).

$$\begin{aligned} t = 0 : \quad \Sigma^1 n^2 &= \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(1+n) + \frac{\delta^2 t^2}{2} \frac{n(1+3n+2n^2)}{6}, \\ t = 1 : \quad \Sigma^1 n^2 &= \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(n-1) + \frac{\delta^2 t^2}{2} \frac{n(1-3n+2n^2)}{6}, \\ t = 2 : \quad \Sigma^1 n^2 &= \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(n-3) + \frac{\delta^2 t^2}{2} \frac{n(13-9n+2n^2)}{6}, \\ t = 3 : \quad \Sigma^1 n^2 &= \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(n-5) + \frac{\delta^2 t^2}{2} \frac{n(37-15n+2n^2)}{6}, \\ t = 4 : \quad \Sigma^1 n^2 &= \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(n-7) + \frac{\delta^2 t^2}{2} \frac{n(73-21n+2n^2)}{6}. \end{aligned}$$

Example 5.5 (Sums of cubes in central differences).

$$\begin{aligned}
t = 0 : \quad \Sigma^1 n^3 &= \frac{t^3}{2}(2n) + \frac{\delta t^3}{2} n(1+n) + \frac{\delta^2 t^3}{2} \frac{n(1+3n+2n^2)}{6} + \frac{\delta^3 t^3}{2} \frac{n(-1+n+4n^2+2n^3)}{24}, \\
t = 1 : \quad \Sigma^1 n^3 &= \frac{t^3}{2}(2n) + \frac{\delta t^3}{2} n(n-1) + \frac{\delta^2 t^3}{2} \frac{n(1-3n+2n^2)}{6} + \frac{\delta^3 t^3}{2} \frac{n(1+n-4n^2+2n^3)}{24}, \\
t = 2 : \quad \Sigma^1 n^3 &= \frac{t^3}{2}(2n) + \frac{\delta t^3}{2} n(n-3) + \frac{\delta^2 t^3}{2} \frac{n(13-9n+2n^2)}{6} + \frac{\delta^3 t^3}{2} \frac{n(-21+25n-12n^2+2n^3)}{24}, \\
t = 3 : \quad \Sigma^1 n^3 &= \frac{t^3}{2}(2n) + \frac{\delta t^3}{2} n(n-5) + \frac{\delta^2 t^3}{2} \frac{n(37-15n+2n^2)}{6} + \frac{\delta^3 t^3}{2} \frac{n(-115+73n-20n^2+2n^3)}{24}, \\
t = 4 : \quad \Sigma^1 n^3 &= \frac{t^3}{2}(2n) + \frac{\delta t^3}{2} n(n-7) + \frac{\delta^2 t^3}{2} \frac{n(73-21n+2n^2)}{6} + \frac{\delta^3 t^3}{2} \frac{n(-329+145n-28n^2+2n^3)}{24}.
\end{aligned}$$

6. REMARK ABOUT CENTRAL FACTORIAL NUMBERS OF THE SECOND KIND

7. GENERALIZED CENTRAL FACTORIAL NUMBERS OF THE SECOND KIND

Definition 7.1 (Generalized Central factorial numbers of the second kind). *For integers $0 \leq k \leq m$, and an arbitrary integer t ,*

$$T_t(m, k) = \frac{\delta^k t^m}{k!},$$

defined by central Newton's formula for powers, evaluated at arbitrary integer t

$$n^m = \sum_{k=0}^{\infty} T_t(m, k)(n-t)^{[k]}.$$

Proposition 7.2 (Generalized Central factorial sums). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^m \left[\binom{n-t+\frac{k}{2}+1}{k+1} - \binom{1-t+\frac{k}{2}}{k+1} + \binom{n-t+\frac{k}{2}}{k+1} - \binom{-t+\frac{k}{2}}{k+1} \right] k! T_t(m, k)$$

n/k	0	1	2	3	4	5	6	7	8
0	1								
1	0	1							
2	0	0	1						
3	0	$\frac{1}{4}$	0	1					
4	0	0	1	0	1				
5	0	$\frac{1}{16}$	0	$\frac{5}{2}$	0	1			
6	0	0	1	0	5	0	1		
7	0	$\frac{1}{64}$	0	$\frac{91}{16}$	0	$\frac{35}{4}$	0	1	
8	0	0	1	0	21	0	14	0	1

Table 1. Values of $T_0(n, k)$ for $0 \leq k \leq n \leq 8$.

n/k	0	1	2	3	4	5	6	7	8
0	1								
1	1	1							
2	1	2	1						
3	1	$\frac{13}{4}$	3	1					
4	1	5	7	4	1				
5	1	$\frac{121}{16}$	15	$\frac{25}{2}$	5	1			
6	1	$\frac{91}{8}$	31	35	20	6	1		
7	1	$\frac{1093}{64}$	63	$\frac{1491}{16}$	70	$\frac{119}{4}$	7	1	
8	1	$\frac{205}{8}$	127	$\frac{483}{2}$	231	126	42	8	1

Table 2. Values of $T_1(n, k)$ for $0 \leq k \leq n \leq 8$.

n/k	0	1	2	3	4	5	6	7	8
0	1								
1	2	1							
2	4	4	1						
3	8	$\frac{49}{4}$	6	1					
4	16	34	25	8	1				
5	32	$\frac{1441}{16}$	90	$\frac{85}{2}$	10	1			
6	64	$\frac{931}{4}$	301	190	65	12	1		
7	128	$\frac{37969}{64}$	966	$\frac{12411}{16}$	350	$\frac{371}{4}$	14	1	
8	256	$\frac{6001}{4}$	3025	3003	1701	588	126	16	1

Table 3. Values of $T_2(n, k)$ for $0 \leq k \leq n \leq 8$.

n/k	0	1	2	3	4	5	6	7	8
0	1								
1	3	1							
2	9	6	1						
3	27	$\frac{109}{4}$	9	1					
4	81	111	55	12	1				
5	243	$\frac{6841}{16}$	285	$\frac{185}{2}$	15	1			
6	729	$\frac{12753}{8}$	1351	585	140	18	1		
7	2187	$\frac{372709}{64}$	6069	$\frac{53011}{16}$	1050	$\frac{791}{4}$	21	1	
8	6561	$\frac{167943}{8}$	26335	$\frac{35049}{2}$	6951	1722	266	24	1

Table 4. Values of $T_3(n, k)$ for $0 \leq k \leq n \leq 8$.

8. DOUBLE AND TRIPLE SUMS OF POWERS IN TERMS OF CENTRAL DIFFERENCES

Proposition 8.1 (Double sums of powers decomposition). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\begin{aligned} \Sigma^2 n^m = \frac{1}{2} \sum_{k=0}^m \left[+ \binom{n-t+\frac{k}{2}+2}{k+2} - \binom{2-t+\frac{k}{2}}{k+2} \Sigma^0 n^0 - \binom{1-t+\frac{k}{2}}{k+1} \Sigma^1 n^0 \right. \\ \left. + \binom{n-t+\frac{k}{2}+1}{k+2} - \binom{1-t+\frac{k}{2}}{k+2} \Sigma^0 n^0 - \binom{0-t+\frac{k}{2}}{k+1} \Sigma^1 n^0 \right] \delta^k t^m \end{aligned}$$

Proposition 8.2 (Triple sums of powers decomposition). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\begin{aligned} \Sigma^3 n^m = \frac{1}{2} \sum_{k=0}^m \left[\right. \\ + \binom{n-t+\frac{k}{2}+3}{k+3} - \binom{3-t+\frac{k}{2}}{k+3} \Sigma^0 n^0 - \binom{2-t+\frac{k}{2}}{k+2} \Sigma^1 n^0 - \binom{1-t+\frac{k}{2}}{k+1} \Sigma^2 n^0 \\ \left. + \binom{n-t+\frac{k}{2}+2}{k+3} - \binom{2-t+\frac{k}{2}}{k+3} \Sigma^0 n^0 - \binom{1-t+\frac{k}{2}}{k+2} \Sigma^1 n^0 - \binom{0-t+\frac{k}{2}}{k+1} \Sigma^2 n^0 \right] \delta^k t^m \end{aligned}$$

Example 8.3 (Double sum of squares in central differences).

$$\begin{aligned} t=0: \quad \Sigma^2 n^2 &= \frac{t^2}{2} n(1+n) + \frac{\delta t^2}{2} \frac{n(2+3n+n^2)}{3} + \frac{\delta^2 t^2}{2} \frac{n(1+n)^2(2+n)}{12}, \\ t=1: \quad \Sigma^2 n^2 &= \frac{t^2}{2} n(1+n) + \frac{\delta t^2}{2} \frac{n(-1+n^2)}{3} + \frac{\delta^2 t^2}{2} \frac{n^2(-1+n^2)}{12}, \\ t=2: \quad \Sigma^2 n^2 &= \frac{t^2}{2} n(1+n) + \frac{\delta t^2}{2} \frac{n(-4-3n+n^2)}{3} + \frac{\delta^2 t^2}{2} \frac{n(10+5n-4n^2+n^3)}{12}, \\ t=3: \quad \Sigma^2 n^2 &= \frac{t^2}{2} n(1+n) + \frac{\delta t^2}{2} \frac{n(-7-6n+n^2)}{3} + \frac{\delta^2 t^2}{2} \frac{n(32+23n-8n^2+n^3)}{12}, \\ t=4: \quad \Sigma^2 n^2 &= \frac{t^2}{2} n(1+n) + \frac{\delta t^2}{2} \frac{n(-10-9n+n^2)}{3} + \frac{\delta^2 t^2}{2} \frac{n(66+53n-12n^2+n^3)}{12}. \end{aligned}$$

Example 8.4 (Triple sum of squares in central differences).

$$\begin{aligned} t=0: \quad \Sigma^3 n^2 &= \frac{t^2}{2} \frac{n(2+3n+n^2)}{3} + \frac{\delta t^2}{2} \frac{n(6+11n+6n^2+n^3)}{12} + \frac{\delta^2 t^2}{2} \frac{n(18+45n+40n^2+15n^3+2n^4)}{120}, \\ t=1: \quad \Sigma^3 n^2 &= \frac{t^2}{2} \frac{n(2+3n+n^2)}{3} + \frac{\delta t^2}{2} \frac{n(-2-n+2n^2+n^3)}{12} + \frac{\delta^2 t^2}{2} \frac{n(-2-5n+5n^3+2n^4)}{120}, \\ t=2: \quad \Sigma^3 n^2 &= \frac{t^2}{2} \frac{n(2+3n+n^2)}{3} + \frac{\delta t^2}{2} \frac{n(-10-13n-2n^2+n^3)}{12} + \frac{\delta^2 t^2}{2} \frac{n(58+65n-5n^3+2n^4)}{120}, \\ t=3: \quad \Sigma^3 n^2 &= \frac{t^2}{2} \frac{n(2+3n+n^2)}{3} + \frac{\delta t^2}{2} \frac{n(-18-25n-6n^2+n^3)}{12} + \frac{\delta^2 t^2}{2} \frac{n(198+255n+40n^2-15n^3+2n^4)}{120}, \\ t=4: \quad \Sigma^3 n^2 &= \frac{t^2}{2} \frac{n(2+3n+n^2)}{3} + \frac{\delta t^2}{2} \frac{n(-26-37n-10n^2+n^3)}{12} + \frac{\delta^2 t^2}{2} \frac{n(418+565n+120n^2-25n^3+2n^4)}{120}. \end{aligned}$$

9. MULTIFOLD SUMS OF POWERS IN TERMS OF CENTRAL DIFFERENCES

Proposition 9.1 (Multifold sums of powers). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\begin{aligned} \Sigma^r n^m = & \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r} \right. \\ & \left. - \sum_{s=0}^{r-1} \left[\binom{r-s-t+\frac{k}{2}}{k+r-s} + \binom{r-1-s-t+\frac{k}{2}}{k+r-s} \right] \Sigma^s n^0 \right\} \delta^k t^m \end{aligned}$$

Example 9.2 (Examples of multifold binomial sums).

$$\begin{aligned} t = 0 : \quad \Sigma^r n^m = & \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n+\frac{k}{2}+r}{k+r} + \binom{n+\frac{k}{2}+r-1}{k+r} \right. \\ & \left. - \sum_{s=0}^{r-1} \left[\binom{r-s+\frac{k}{2}}{k+r-s} + \binom{r-1-s+\frac{k}{2}}{k+r-s} \right] \binom{s+n-1}{s} \right\} \delta^k 0^m, \\ t = 1 : \quad \Sigma^r n^m = & \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n-1+\frac{k}{2}+r}{k+r} + \binom{n-1+\frac{k}{2}+r-1}{k+r} \right. \\ & \left. - \sum_{s=0}^{r-1} \left[\binom{r-s-1+\frac{k}{2}}{k+r-s} + \binom{r-2-s+\frac{k}{2}}{k+r-s} \right] \binom{s+n-1}{s} \right\} \delta^k 1^m, \\ t = 2 : \quad \Sigma^r n^m = & \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n-2+\frac{k}{2}+r}{k+r} + \binom{n-2+\frac{k}{2}+r-1}{k+r} \right. \\ & \left. - \sum_{s=0}^{r-1} \left[\binom{r-s-2+\frac{k}{2}}{k+r-s} + \binom{r-3-s+\frac{k}{2}}{k+r-s} \right] \binom{s+n-1}{s} \right\} \delta^k 2^m. \end{aligned}$$

10. MULTIFOLD SUMS OF POWERS IN GENERALIZED CENTRAL FACTORIAL NUMBERS

Proposition 10.1 (Multifold central factorial sums of powers). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\begin{aligned} \Sigma^r n^m = & \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r} \right. \\ & \left. - \sum_{s=0}^{r-1} \left[\binom{r-s-t+\frac{k}{2}}{k+r-s} + \binom{r-1-s-t+\frac{k}{2}}{k+r-s} \right] \binom{s+n-1}{s} \right\} k! T_t(m, k) \end{aligned}$$

t	$T_t(2m, 2k)$	OEIS ID
0	1, 0, 1, 0, 1, 1, 0, 1, 5, 1, 0, 1, 21, 14, 1, 0, 1, 85, 147, 30, 1, $\dots$	A269945
1	1, 1, 1, 1, 7, 1, 1, 31, 20, 1, 1, 127, 231, 42, 1, 1, 511, $\dots$	A394692
2	1, 4, 1, 16, 25, 1, 64, 301, 65, 1, 256, 3025, 1701, 126, 1, $\dots$	A395456
3	1, 9, 1, 81, 55, 1, 729, 1351, 140, 1, 6561, 26335, 6951, 266, 1, $\dots$	A395457

Example 10.2 (Sums of squares in central factorial numbers).

$$t = 0 : \quad \Sigma^1 n^2 = n \cdot 0 + \frac{1}{2}n(1+n) \cdot 0 + \frac{1}{6}n(1+3n+2n^2) \cdot 1,$$

$$t = 1 : \quad \Sigma^1 n^2 = n \cdot 1 + \frac{1}{2}n(n-1) \cdot 2 + \frac{1}{6}n(1-3n+2n^2) \cdot 1,$$

$$t = 2 : \quad \Sigma^1 n^2 = n \cdot 4 + \frac{1}{2}n(n-3) \cdot 4 + \frac{1}{6}n(13-9n+2n^2) \cdot 1,$$

$$t = 3 : \quad \Sigma^1 n^2 = n \cdot 9 + \frac{1}{2}n(n-5) \cdot 6 + \frac{1}{6}n(37-15n+2n^2) \cdot 1,$$

$$t = 4 : \quad \Sigma^1 n^2 = n \cdot 16 + \frac{1}{2}n(n-7) \cdot 8 + \frac{1}{6}n(73-21n+2n^2) \cdot 1,$$

$$t = 5 : \quad \Sigma^1 n^2 = n \cdot 25 + \frac{1}{2}n(n-9) \cdot 10 + \frac{1}{6}n(121-27n+2n^2) \cdot 1.$$

Example 10.3 (Sums of cubes in central factorial numbers).

$$\begin{aligned}
t = 0 : \quad \Sigma^1 n^4 &= n \cdot 0 + \frac{1}{2}n(1+n) \cdot 0 + \frac{1}{6}n(1+3n+2n^2) \cdot 1 \\
&\quad + \frac{1}{8}n(-1+n+4n^2+2n^3) \cdot 0 + \frac{1}{10}n(-2-5n+5n^3+2n^4) \cdot 1, \\
t = 1 : \quad \Sigma^1 n^4 &= n \cdot 1 + \frac{1}{2}n(n-1) \cdot 5 + \frac{1}{6}n(1-3n+2n^2) \cdot 7 \\
&\quad + \frac{1}{8}n(1+n-4n^2+2n^3) \cdot 4 + \frac{1}{10}n(-2+5n-5n^3+2n^4) \cdot 1, \\
t = 2 : \quad \Sigma^1 n^4 &= n \cdot 16 + \frac{1}{2}n(n-3) \cdot 34 + \frac{1}{6}n(13-9n+2n^2) \cdot 25 \\
&\quad + \frac{1}{8}n(-21+25n-12n^2+2n^3) \cdot 8 + \frac{1}{10}n(18-45n+40n^2-15n^3+2n^4) \cdot 1, \\
t = 3 : \quad \Sigma^1 n^4 &= n \cdot 81 + \frac{1}{2}n(n-5) \cdot 111 + \frac{1}{6}n(37-15n+2n^2) \cdot 55 \\
&\quad + \frac{1}{8}n(-115+73n-20n^2+2n^3) \cdot 12 + \frac{1}{10}n(298-275n+120n^2-25n^3+2n^4) \cdot 1, \\
t = 4 : \quad \Sigma^1 n^4 &= n \cdot 256 + \frac{1}{2}n(n-7) \cdot 260 + \frac{1}{6}n(73-21n+2n^2) \cdot 97 \\
&\quad + \frac{1}{8}n(-329+145n-28n^2+2n^3) \cdot 16 + \frac{1}{10}n(1318-805n+240n^2-35n^3+2n^4) \cdot 1, \\
t = 5 : \quad \Sigma^1 n^4 &= n \cdot 625 + \frac{1}{2}n(n-9) \cdot 505 + \frac{1}{6}n(121-27n+2n^2) \cdot 151 \\
&\quad + \frac{1}{8}n(-711+241n-36n^2+2n^3) \cdot 20 + \frac{1}{10}n(3798-1755n+400n^2-45n^3+2n^4) \cdot 1.
\end{aligned}$$

11. MULTIFOLD SUMS OF POWERS IN BINOMIAL FORM

Lemma 11.1 (Multifold sum of zero powers). *For integers $r \geq 0$ and $n \geq 1$*

$$\Sigma^r n^0 = \binom{r+n-1}{r}$$

Proof. (1) Let $r = 0$, then we have $\Sigma^0 n^0 = 1$, by definition (1.1).

(2) Let $r = 1$, then we have $\Sigma^1 n^0 = \sum_{k=1}^n 1 = \binom{n}{1}$.

(3) Let $r = 2$, then we have $\Sigma^2 n^0 = \sum_{k=1}^n \binom{k}{1} = \binom{n+1}{2}$.

(4) Let $r = 3$, then we have $\Sigma^3 n^0 = \sum_{k=1}^n \binom{k+1}{2} = \binom{n+2}{3}$.

(5) Similarly, for arbitrary r , by hockey stick identity $\sum_{k=1}^n \binom{k}{r} = \binom{n+1}{r+1}$, thus, the claim follows $\Sigma^r n^0 = \sum_{k=1}^n \Sigma^{r-1} k^0 = \sum_{k=1}^n \binom{k+r-2}{r-1} = \binom{r+n-1}{r}$.

□

Proposition 11.2 (Multifold binomial sums of powers). *For integers $n, m \geq 0$, and an arbitrary integer t ,*

$$\begin{aligned} \Sigma^r n^m = & \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r} \right. \\ & \left. - \sum_{s=0}^{r-1} \left[\binom{r-s-t+\frac{k}{2}}{k+r-s} + \binom{r-1-s-t+\frac{k}{2}}{k+r-s} \right] \binom{s+n-1}{s} \right\} \delta^k t^m \end{aligned}$$

12. RELATED WORKS

13. FRAMEWORK FOR SUMS OF POWERS

14. CONCLUSIONS

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15. MATHEMATICA PROGRAMS

Use this [GitHub](#) repository to validate the results using Mathematica programs.

Mathematica Function	Validates / Prints
<code>MultifoldSumOfPowersRecurrence[r, n, m]</code>	Computes $\Sigma^r n^m$, (1.1)
<code>CentralFactorial[n, k]</code>	Computes $n^{[k]}$, (1.2)
<code>ValidateHalvedCentralBinomials[20]</code>	Validates Lem. (4.2)
<code>ValidateNewtonsFormulaForPowers[10]</code>	Validates Prop. (4.3)
<code>ValidateHalvedCentralFactorialPowers[5]</code>	Validates Prop. (4.4)
<code>ValidateCenteredHockeyStickIdentity[5]</code>	Validates Lem. (4.6)
<code>ValidateOrdinarySumsOfPowersDecomposition[10]</code>	Validates Prop. (5.2)
<code>ValidateDoubleSumsOfPowersDecomposition[5]</code>	Validates Prop. (8.1)
<code>ValidateTripleSumsOfPowersDecomposition[5]</code>	Validates Prop. (8.2)
<code>ValidateMultifoldSumsOfPowers[3]</code>	Validates Prop. (9.1)